

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 4

Duration : 1 hour

Total Marks:40

Annexure A

ARTICLE REVIEW

Code of Conduct:

I HEMASRINITHA P (192572179) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Hemasrinitha P

ARTICLE REVIEW

INSTRUCTIONS

Select any ONE peer-reviewed research article (published after 2019) from IEEE Xplore, Springer, Elsevier, or ACM Digital Library related to any of the following topics: *Inductive Learning, Decision Trees, Statistical Learning Methods (Naïve Bayes / Logistic Regression / SVM), or Reinforcement Learning applications.*

Submit: (i) Article citation details (ii) PDF or valid DOI link (iii) Written review as per the tasks below.

Task 1: Article Summary

(a) Provide the full citation of the article (Author(s), Title, Journal/Conference, Year, DOI). State the domain/application area and the specific ML problem addressed.

(b) Summarise the key objective of the article in your own words (150–200 words). Clearly state the research gap the authors identified and how they proposed to fill it.

Task 2: Methodology Analysis

(a) Explain the ML technique(s) used in the article (e.g., Decision Tree variant, RL algorithm, Statistical model). Describe the dataset used—size, features, source, and how it was prepared.

(b) Map the technique to CO 4 topics (Inductive Learning / Decision Trees / Statistical Learning / RL). Identify one methodological strength and one limitation in the authors' approach.

Task 3: Results & Critical Evaluation

(a) Report the key results (accuracy, F1-Score, AUC-ROC, reward convergence, etc.) as presented by the authors. Create a comparative table if multiple models are benchmarked.

(b) Critically evaluate the results: Are the reported metrics realistic? Are there potential biases (dataset, evaluation protocol)? What additional experiments would strengthen the findings?

(c) Discuss real-world applicability: Where can this technique be deployed? What are the challenges in scaling the solution to industry (data availability, compute, ethical issues)?

Task 4: Reflection & Learning Outcome

(a) State three key insights you gained from reading the article that deepen your understanding of CO 4 topics. Connect each insight to a specific concept from the course syllabus.

(b) If you were to extend this research, propose one novel experiment or enhancement (different dataset / improved algorithm / new application domain). Justify its feasibility and expected impact.

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Article	20%	Demonstrates clear and in-depth understanding of the article's purpose, concepts, and arguments	Shows good understanding with minor gaps	Partial understanding; misses key ideas	Lacks understanding of the article
Summary	20%	Provides concise and accurate summary highlighting all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Provides insightful critique with strong reasoning and evaluation	Provides reasonable analysis with some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples, data, or references to support review	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear, and coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganized and difficult to understand

TASK 1: ARTICLE SUMMARY

(a) Article Citation Details

Selected Article: Lynam, A. L., Dennis, J. M., Owen, K. R., et al. "Logistic regression has similar performance to optimised machine learning algorithms in a clinical setting: application to the discrimination between type 1 and type 2 diabetes in young adults." *Diagnostic and Prognostic Research*, Volume 4, Article 6, 2020.

DOI: 10.1186/s41512-020-00075-2

DOI / Full Article

Domain: Healthcare / Medical Diagnosis

ML Problem: Classification of diabetes type, specifically distinguishing Type 1 diabetes from Type 2 diabetes using patient characteristics.

The study compared Logistic Regression with six machine-learning methods: Gradient Boosting Machine, MARS, Neural Network, K-Nearest Neighbours, Random Forest and Support Vector Machine.

(b) Key Objective and Research Gap

The main objective of the article was to determine whether a traditional statistical method, Logistic Regression, could perform as well as more advanced machine-learning algorithms for distinguishing Type 1 diabetes from Type 2 diabetes. The researchers used seven clinical predictors, including age, BMI, sex, cholesterol-related measurements and diabetes-related biomarkers. They trained and optimised seven different classification models and evaluated them using an independent external dataset.

The research gap identified by the authors was that although many machine-learning techniques were increasingly being used in medical prediction, there was a need to determine whether these more complex approaches actually provided better clinical performance than simpler and well-established statistical models. The study addressed this gap by directly comparing Logistic Regression with several machine-learning algorithms using the same clinical data. The results showed that Logistic Regression performed very well, achieving an external ROC-AUC of 0.95 and performing similarly to the more complex models. The authors therefore highlighted that when a small number of strong and well-understood predictors are available, a simpler statistical model can be an effective and clinically useful choice.

TASK 2: METHODOLOGY ANALYSIS

(a) ML Techniques and Dataset

The article mainly uses Logistic Regression as a statistical learning method and compares it with six machine-learning techniques: Gradient Boosting Machine, Multivariate Adaptive Regression Splines (MARS), Neural Network, K-Nearest Neighbours (KNN), Random Forest, and Support Vector Machine (SVM). Logistic Regression was used to estimate the probability that a young adult had Type 1 rather than Type 2 diabetes. The models were trained and tuned using the same clinical predictors and their performance was compared using ROC-AUC.

The study used data from young adults with diabetes collected through the ADDRESS-2 study, with additional external validation using data from the Young Diabetes in Oxford (YDX) study. The main predictors included age at diagnosis, BMI, sex, and diabetes-related biomarkers, including glucose, HDL cholesterol and other clinical measurements. The data were prepared by selecting relevant clinical variables, handling missing values, and standardising the predictors where required for the machine-learning methods. The researchers used an independent external dataset to assess how well the models generalised to new patients.

(b) Mapping to CO 4 Topics

The article mainly relates to Statistical Learning, because Logistic Regression is a statistical classification technique. It also relates to Inductive Learning, as the models learn patterns from existing labelled patient records and use them to classify new patients.

Methodological strength: A major strength is the comparison of Logistic Regression with several different machine-learning methods using the same clinical data. The use of an independent external dataset for validation also provides stronger evidence that the results can generalise beyond the training data.

Methodological limitation: One limitation is that the study focuses on a specific young-adult diabetes population, so the results may not directly generalise to older patients or populations with different clinical characteristics. Also, the number of predictors was relatively limited, which may restrict the ability of more complex models to demonstrate an advantage over Logistic Regression.

TASK 3: RESULTS & CRITICAL EVALUATION

(a) Key Results

The authors compared Logistic Regression with six other machine-learning models. They used ROC-AUC as the main measure of classification performance. The study did not report a single accuracy or F1-score as the main comparison measure.

Model	Internal Validation AUC	External Validation AUC
Gradient Boosting Machine	0.96	0.93
K-Nearest Neighbours	0.93	0.92
Logistic Regression	0.96	0.95
MARS	0.96	0.94
Neural Network	0.96	0.94
Random Forest	0.95	0.94
Support Vector Machine	0.96	0.94

The results show that all models performed well, with internal AUC values of at least 0.93. On the independent external dataset, performance decreased slightly, but all models still achieved AUC values of at least 0.92. Logistic Regression achieved the highest external AUC of **0.95**. The Neural Network and Gradient Boosting Machine showed the best calibration performance.

(b) Critical Evaluation

The reported AUC values are strong, but they are reasonable because the study used a small number of strong clinical predictors. The external validation results are particularly important because they show that performance remained high on a different dataset.

However, there are some limitations. The development dataset contained 960 participants, with only 14% having Type 1 diabetes, and the external dataset contained 504 participants, with 21% having Type 1 diabetes. The study population was also limited to adults aged 18–50 years, so the results may not generalise to older patients or other populations.

The authors already used nested cross-validation and external validation, which strengthens the evaluation. Further studies could include larger and more diverse datasets, different countries and

healthcare systems, longer-term validation, and prospective clinical testing. Testing the models on more demographic groups would also help identify possible bias.

(c) Real-World Applicability

The technique could be used in hospitals, diabetes clinics and clinical decision-support systems to help distinguish Type 1 and Type 2 diabetes. Logistic Regression is particularly suitable because it is simple, relatively easy to interpret, and achieved performance similar to the more complex models.

For industry deployment, important challenges include:

- **Data availability:** Large, high-quality and representative patient datasets are required.
- **Data privacy:** Patient information must be securely stored and handled.
- **Generalisation:** A model trained on one population may not perform equally well elsewhere.
- **Clinical validation:** Models should be tested prospectively before being used for real patient decisions.
- **Explainability:** Doctors should be able to understand and evaluate the model's predictions.
- **Bias and fairness:** The system should be checked for unequal performance across different patient groups.

Overall, the study demonstrates that a simple Logistic Regression model can be a practical clinical solution when a small number of strong predictors are available, rather than automatically choosing a more complex machine-learning model.

TASK 4: REFLECTION & LEARNING OUTCOME

(a) Key Insights

1. Simple models can perform very well

The article showed that Logistic Regression achieved performance similar to more complex machine-learning models. This helped me understand that a complex model is not always the best choice.

CO 4 Concept: Statistical Learning – Logistic Regression is a statistical learning technique used for classification and probability prediction.

2. Learning from existing data is the basis of inductive learning

The models learned patterns from previously collected patient data and used those patterns to classify new patients. This helped me understand how general rules can be learned from examples.

CO 4 Concept: Inductive Learning – The model learns a general relationship from labelled training examples.

3. External validation is important

The study tested the models using an independent dataset. The performance was slightly different from the internal validation results, showing that a model must be tested on unseen data before real-world use.

CO 4 Concept: Statistical Model Evaluation – A model should be evaluated using appropriate measures such as ROC-AUC and validated on independent data.

(b) Proposed Extension

I would extend the research by testing the models on a larger and more diverse international diabetes dataset containing patients from different age groups, ethnic backgrounds and healthcare settings.

The same Logistic Regression, Random Forest, SVM and other models could be trained and compared using AUC-ROC, precision, recall and F1-score. An explainable AI method could also be added to show which patient features contribute most to each prediction.

This experiment is feasible because publicly available healthcare datasets can be used for initial testing. A larger and more diverse dataset would help determine whether the model performs consistently across different patient populations. The expected impact is a more reliable, fair and generalisable diabetes classification system suitable for future clinical decision-support applications.